

FILM IS DEAD



The age of digital video is dawning and everyone, from independent filmmakers to handycam enthusiasts, are taking note

Two decades ago, the word-processor transformed the world of the writer. No more typos, no more piles of crumpled papers. You could write an entire story, and if you didn't like the protagonist's name, a find and replace string could change it all. You could shuffle those words around, grab a paragraph from down there and move it up here.

Word processors, spreadsheets, image editing software, and the Net have bought about radical changes in the way we work.

Now, the amateur moviemaker gets the big break. Just like shuffling words, he's juggling video clips. No negative cutting, no off-site sound mixing. No elaborate special effects done at huge costs.

Inexpensive broadcast quality equipment has democratised both production and distribution; the mediums of film and television are being redefined on the Internet by a new generation of technologically savvy media artists. The results they achieve are difficult to quantify, but what is clear is that visual media is undergoing a dramatic and fundamental transformation.

Introducing digital video

Endorsed by 55 electronics manufacturers, the new digital video (DV) standard specifies a level of picture quality that is significantly better than what's available today on television. Needless to say, the DV standard is far superior to other

common standards with lower quality, such as the familiar VHS standard.

Until the development of the Digital Video format (and its attendant hardware), you had to add expensive video digitising cards to your computer if you wanted to edit video. After editing, the same hardware used to write the edited footage back onto tape.

With DV, video footage is digitised and transferred to your computer, where scenes can be inserted, deleted, or rearranged in any order. A DV camera digitises video, compresses it, and writes a digital file to its on-board storage—while shooting the video. All you have to do is copy that digital file from the camera's tape onto your computer. When you make copies of your work in the digital language, you don't lose picture quality

like we used to with 'analog'.

But the DV format is more than just a digital version of ordinary analog video. In overall image quality, resolution, colour reproduction and sharpness, DV far outperforms older consumer and industrial formats. In fact, DV delivers image quality that rivals high-end broadcast formats such as Betacam SP.

Enter FireWire

You can make the grooviest video-editing hardware and software in the world, but it's not going to do anything until the raw material is in the machine. So how do you get it there?

Answer: FireWire, a high-speed interface for connecting peripherals to computers. Currently running at 400 MBps, it supports up to 63 devices, is hot pluggable and requires no configuration. The next generation is expected to reach 800 MBps. FireWire was developed by Apple, but has been adopted as an industry standard, IEEE 1394. The name 'FireWire' is a trademark of Apple, however, so other companies give the technology different names, such as Sony's iLink.

DV has a data rate of 25 megabits per second (Mbps), meaning that each second of video requires 25 megabits (roughly 3.1 megabytes) data per second. The standard FireWire interface built into many computers these days can transfer about 12.5 megabytes per second, which allows it to handle the DV format easily. →

To DV or not to DV

Price is a big clincher when opting for a DV setup. A professional setup with non-linear editing software and hardware, a Betaplayer and camera would cost you Rs 20-30 lakh. The DV route, inclusive of a competent PC setup, a camcorder, and software would cost you considerably less. Even if you add another Rs 50,000 for professional video editing software and decide to go for a Mac setup, a DV setup would cost you about Rs 3 to 5 lakh.